

Analysis and Optimization of a New 2-D Magnet Array for Planar Motor

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This paper presents a new 2-D permanent-magnet array for a planar motor, in which the angle between the magnetization directions of any two adjacent magnets is 45° . The harmonic model for flux density distribution of the array is solved by the scalar magnetic potential equation and validated by the finite-element method. An analytical model for real-time control is derived by taking the first harmonic of the magnetic flux density distribution. The ignored higher harmonics in z -component of the magnetic flux density distribution is minimized by the genetic algorithm such that the analytical model becomes more accurate. Compared with the well-known Halbach magnet array, the proposed magnet array has lower higher harmonic components and higher z -component of the magnetic flux density, which will reduce the force ripples of the planar motor.

Index Terms—Force ripples, genetic algorithm, magnet array, planar motor.

I. INTRODUCTION

IN many industrial apparatus, such as pick-and-place machines and inspection systems, objects are positioned and moved in a horizontal plane. The planar motors, which can move freely in the plane, have received increasing attention in the academic and industrial field [1]–[4]. There are three main types of planar motors: induction motors, sawyer motors and synchronous permanent-magnet planar motors (SPMPM) [2]. Synchronous permanent-magnet planar motors can realize large range planar motion with high speed and high precision, and become a research focus for planar motor. Fig. 1 shows the shape of a moving-magnet, magnetically levitated planar motor which is driven by Lorentz force and can be used in extreme-UV lithography equipment [5]. The stator of this motor consists of coil arrays. The moving part is a 2-D permanent-magnet array. Analysis and optimal design of the magnet array is a very important step in the design of planar motors.

To obtain a larger thrust force in the motor, a 2-D magnet array with higher magnetic flux density is needed. Furthermore, in the analysis of the planar motor, especially in the real-time control, a simple and accurate analytical model of the magnet array is also needed to obtain the model of the force and torque in the planar motor. The harmonic model for flux density distribution of some types of 2-D magnet arrays which are presented in literature can be derived using Fourier analysis and the scalar magnetic potential equation [6], [7]. But the harmonic model is too complex to derive the model for planar motor. To deal with this problem, Cho *et al.* [2] simplify the flux density distribution in the air gap to a 2-D sinusoidal wave with the flux density decreases exponentially across the air-gap length. Jansen *et al.* [7], [8] derive an analytical model by only taking the first harmonic of the magnetic flux density distribution and optimize the 2-D Halbach magnet array by maximizing the magnetic flux density per unit of moving-part to get lower power dissipation in the motor. However, the ignored higher harmonics component of the air-gap flux density is usually a main reason for the

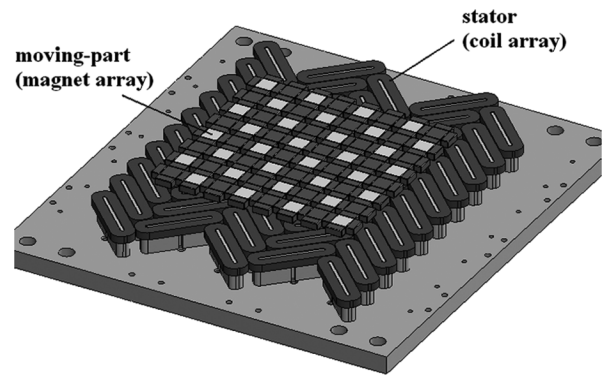


Fig. 1. Scheme of a magnetically levitated planar motor.

force ripples in the planar motor [5], [9]. So a magnet array with higher magnetic flux density and lower higher harmonics components is needed.

Trumper *et al.* [10] proved that there were no higher harmonics components of the scalar magnetic potential on either surface of the ideal 1-D Halbach array, where the magnetization varies continuously with the lateral dimension. Lee *et al.* [9] investigated three magnetizing patterns of 1-D magnet array and found that 45° magnetizing pattern performs better than both the 60° and 90° in a permanent-magnet linear synchronous motor due to decrease of higher harmonics components, which reduces the force ripples of the motor. However, little research has been done on the minimization of the higher harmonics components for a 2-D magnet array. This paper focuses on a new magnet array with lower higher harmonics components. It can be presumed that there were no higher harmonics components of the magnetic flux density distribution of 2-D ideal Halbach array. Although it is not practical to fabricate a 2-D ideal Halbach array to decrease the higher harmonics components. In this paper, a new 2-D magnet array is proposed and the magnetic field distribution is derived by the scalar magnetic potential equation. Afterwards, the higher harmonics in z -component of the magnetic flux density distribution is minimized by the genetic algorithm. In the end, the new array is compared with the Halbach array used by Jansen [5].

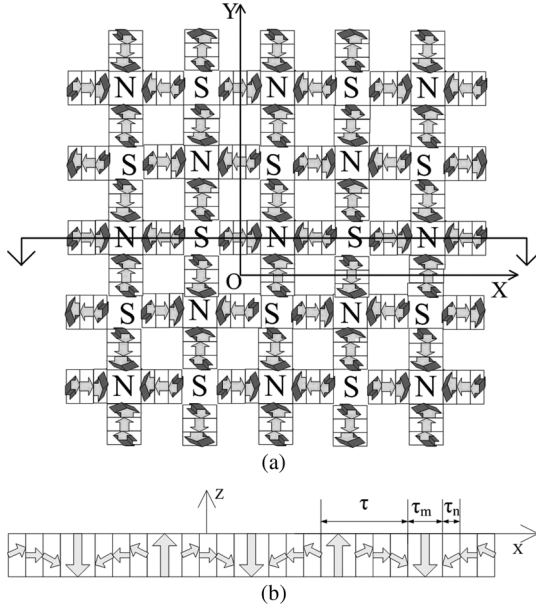


Fig. 2. New magnet array. (a) Top view. (b) Cut-view.

II. ANALYSIS OF AIR-GAP FLUX DENSITY

1) *Analysis Model*: Fig. 2 shows the new magnet array proposed in this paper, including a top view and a cut-view. In Fig. 2, the arrow denotes the magnetization direction of the permanent magnet from the *s*-pole to *n*-pole, where S means the direction toward the paper and N means out of paper. The angle between the magnetization directions of any two adjacent magnets is 45° .

The coordinate system of the new magnet array is shown in Fig. 2. τ is the pole pitch. τ_m is the length of the side of the magnets which are magnetized in *z*-direction. τ_n is the length of the side of the magnets, in which the angle between the magnetization directions and *z*-direction is 45° (see also Fig. 2).

If the edge effects are neglected, the magnet array can be modeled as an infinitely large magnet array [7] and the magnetization vector function *M* describing the array shown in Fig. 2 can be expressed as Fourier series [11], where *k* and *l* are the harmonic numbers for the *x*- and *y*- direction.

$$\vec{M} = \frac{1\text{Br}}{\mu_0} \sum_{k=1}^{\infty} \sum_{l=1}^{\infty} \left[- \left[a(k)b(l) + \sqrt{2}g(k)b(l) \right] \times \cos\left(\frac{k\pi x}{\tau}\right) \sin\left(\frac{l\pi y}{\tau}\right) - \left[b(k)a(l) + \sqrt{2}b(k)g(l) \right] \times \sin\left(\frac{k\pi x}{\tau}\right) \cos\left(\frac{l\pi y}{\tau}\right) \times \left[b(k)b(l) + \sqrt{2}b(k)f(l) + \sqrt{2}b(l)f(k) \right] \times \sin\left(\frac{k\pi x}{\tau}\right) \sin\left(\frac{l\pi y}{\tau}\right) \right] \quad (1)$$

$$a(k) = \frac{4}{k\pi} \cos\left(\frac{k(\tau_m + 2\tau_n)\pi}{2\tau}\right) \sin\left(\frac{k\pi}{2}\right) \quad (2)$$

$$b(k) = \frac{4}{k\pi} \sin\left(\frac{k\tau_m\pi}{2\tau}\right) \sin\left(\frac{k\pi}{2}\right) \quad (3)$$

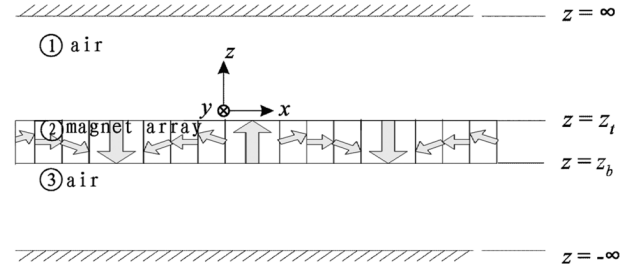


Fig. 3. Space containing magnet array divided into three regions.

$$g(k) = \frac{4}{k\pi} \sin\left(\frac{k(\tau_m + \tau_n)\pi}{2\tau}\right) \sin\left(\frac{k\tau_n\pi}{2\tau}\right) \sin\left(\frac{k\pi}{2}\right) \quad (4)$$

$$f(k) = \frac{4}{k\pi} \cos\left(\frac{k(\tau_m + \tau_n)\pi}{2\tau}\right) \sin\left(\frac{k\tau_n\pi}{2\tau}\right) \sin\left(\frac{k\pi}{2}\right). \quad (5)$$

2) *Governing and Solving Equation*: In order to derive the flux density distribution of the new magnet array, we can divide the space containing the array into three regions, which are shown in Fig. 3. Region 1 and 3 are in air. Region 2 contains the permanent magnets, where the empties of the array are neglected. Because modern NdFeB permanent magnets have a relative permeability of 1.03–1.05, which is close to the relative permeability of air, the error caused by the neglect is small.

There is no current in the space of the problem, so the scalar magnetic potential equation can be used. In region 1 and 3, the following equations apply:

$$\nabla^2 \Psi_n = 0 \quad (6)$$

$$\vec{H}_n = -\nabla \Psi_n \quad (7)$$

$$\vec{B}_n = \mu_0 \vec{H}_n \quad (8)$$

$$\nabla \cdot \vec{B}_n = 0 \quad (9)$$

where the subscript denotes the region (*n* take 1 or 3), $\vec{B}$ is the magnetic flux density, $\vec{H}$ is the magnetic field strength, and Ψ is the scalar potential.

In region 2:

$$\nabla^2 \Psi_2 = \nabla \cdot \frac{\vec{M}}{\mu_r} \quad (10)$$

$$\vec{H}_2 = -\nabla \Psi_2 \quad (11)$$

$$\nabla \times \vec{H}_2 = 0 \quad (12)$$

$$\vec{B}_2 = \mu_0 \mu_r \vec{H}_2 + \mu_0 \vec{M} \quad (13)$$

$$\nabla \cdot \vec{B}_2 = 0. \quad (14)$$

The boundary conditions are as follows (see also Fig. 3):

$$\Psi_1(z = \infty) = 0 \quad (15)$$

$$\Psi_3(z = -\infty) = 0 \quad (16)$$

$$H_{1x}(z = z_t) = H_{2x}(z = z_t) \quad (17)$$

$$H_{1y}(z = z_t) = H_{2y}(z = z_t) \quad (18)$$

$$H_{2x}(z = z_b) = H_{3x}(z = z_b) \quad (19)$$

$$H_{2y}(z = z_b) = H_{3y}(z = z_b) \quad (20)$$

$$B_{1z}(z = z_t) = B_{2z}(z = z_t) \quad (21)$$

$$B_{2z}(z = z_b) = B_{3z}(z = z_b). \quad (22)$$

TABLE I
OPTIMIZATION PARAMETERS

Optimization parameter	Symbol	Value	Unit
Pole pitch	τ	0.025	m
Position of the top surface of the array	Zt	0	m
Position of the bottom surface of the array	Zb	-0.007	m
Remanence of the permanent magnets	Br	1.24	T

TABLE II
OPTIMIZATION VARIABLES OF THE NEW MAGNET ARRAY

Optimization variables	constraints
τ_m/τ	[0,1]
τ_n/τ	[0,(1- τ_m/τ)/2]

Using all the equations above, the flux density distribution can be obtained by separation of variables [7], [12], [13]. The resulting expression for the magnetic flux density distribution in the region 3 is as follows:

$$\vec{B}_3(x, y, z) = -\mu_0 \mu_r \sum_{k=1}^{\infty} \sum_{l=1}^{\infty} K e^{\lambda z} \begin{bmatrix} \frac{k\pi}{\tau} \cos\left(\frac{k\pi x}{\tau}\right) \sin\left(\frac{l\pi y}{\tau}\right) \\ \frac{l\pi}{\tau} \sin\left(\frac{k\pi x}{\tau}\right) \cos\left(\frac{l\pi y}{\tau}\right) \\ \lambda \sin\left(\frac{k\pi x}{\tau}\right) \sin\left(\frac{l\pi y}{\tau}\right) \end{bmatrix} \quad (23)$$

when $\mu_r = 1$,

$$\begin{aligned} K &= B_r(e^{-\lambda z_t} - e^{-\lambda z_b}) \\ &\bullet \frac{[b(k)b(l) + \sqrt{2}b(k)f(l) + \sqrt{2}b(l)f(k)] \pi(k^2 + l^2)}{2(k^2 + l^2)\pi\lambda\mu_0} \\ &+ B_r(e^{-\lambda z_t} - e^{-\lambda z_b}) \\ &\bullet \frac{[a(k)b(l) + \sqrt{2}g(k)b(l)] k\lambda\tau + [b(k)a(l) + \sqrt{2}b(k)g(l)] l\lambda\tau}{2(k^2 + l^2)\pi\lambda\mu_0} \end{aligned} \quad (24)$$

III. OPTIMIZATION OF THE NEW MAGNET ARRAY

Considering the thrusts of the planar motor in x -direction and y -direction are generated by z -component of the air-gap flux density of the magnet array, the objective of optimization is minimization of higher harmonics components of the z -component. The optimization parameters and variables, which are indicated in Figs. 2 and 3, are listed in Tables I and II.

In order to meet the rapidity of real-time control for planar motor, an analytical model of z -component of the magnetic flux density distribution in region 3 is obtained by only taking the first harmonic of z -component from (23).

$$B_{3Z}^{1,1}(x, y, z) = -\mu_0 \mu_r K e^{\lambda z} \left(\lambda \sin\left(\frac{\pi x}{\tau}\right) \sin\left(\frac{\pi y}{\tau}\right) \right). \quad (25)$$

The superscript of magnetic flux density $\vec{B}$ denotes the harmonic numbers for the x - and y -direction and the subscript of magnetic flux density $\vec{B}$ denotes z -component in (25).

The higher harmonics components of the new magnet array in point P(x, y, z) can be evaluated by

$$B_{3Zh}(x, y, z) = B_{3Z}(x, y, z) - B_{3Z}^{1,1}(x, y, z). \quad (26)$$

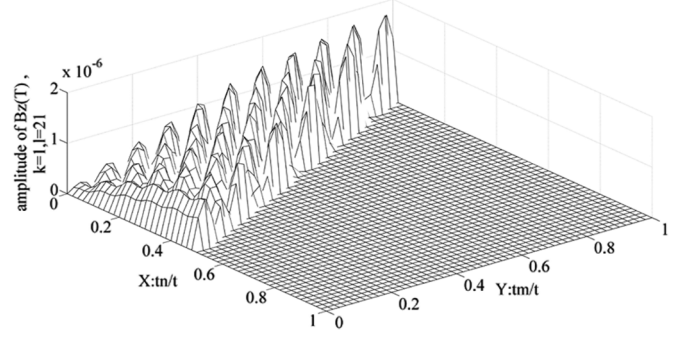


Fig. 4. The amplitude of z -component when $k = 1, l = 21$.

The amplitude of z -component of the magnetic flux density distribution can be predicted with (23) by Matlab. If $k = 1, l = 21$ and the optimization variables varies under the constraints shown in Table II, it is found that the amplitude of the z -component was less than 2×10^{-6} T (see in Fig. 4), which is much less than geomagnetic field (about 6×10^{-5} T). So the harmonic components can be ignored in optimization if any one of the harmonic numbers (k or l) is more than 19. Consequently, we can get an approximate expression of the magnetic flux density distribution for optimization in the region 3. The approximate expression is as follows:

$$B_{3Z}(x, y, z) = -\mu_0 \mu_r \sum_{k=1}^{19} \sum_{l=1}^{19} K e^{\lambda z} \times \left(\lambda \sin\left(\frac{k\pi x}{\tau}\right) \sin\left(\frac{l\pi y}{\tau}\right) \right). \quad (27)$$

The higher harmonics components' effect of z -component of the magnetic flux density in an area of which both the width and length are a half pitch is chosen as the objectiveness of minimization. The area has a rectangle shape and is placed in the plan 4 mm below the bottom of the array. In order to enable a short optimization time while pursuing accuracy, a 25×25 point matrix in the area is taken. Finally, the sum of evaluations for higher harmonics components of all the points is used as the objectiveness of minimization and the following objective function can be obtained:

$$f(\tau_m/\tau, \tau_n/\tau) = \frac{\left| \sum_{m=1}^{25} \sum_{n=1}^{25} B_{3Zh}(P_{mn}) \right|}{\left| \sum_{m=1}^{25} \sum_{n=1}^{25} B_{3Z}^{1,1}(P_{mn}) \right|} \times 100\% \quad (28)$$

$$\text{where : } P_{mn}(0.02m\tau, 0.02n\tau, -0.011). \quad (29)$$

The optimization of (28) has been carried out with genetic algorithm toolbox in Matlab. The constraints are shown as Table II. The population size is 100, and the maximum generation is 100 [14]. The other parameters use the default setup. The genetic algorithm iteration process is revealed in Fig. 5. As a result, the minimization of the objective function can be obtained if τ_m/τ takes 0.4343 and τ_n/τ takes 0.2039.

Fig. 6 shows air-gap flux density distributions of the proposed magnet array on the x - y plane which is 4 mm apart from the

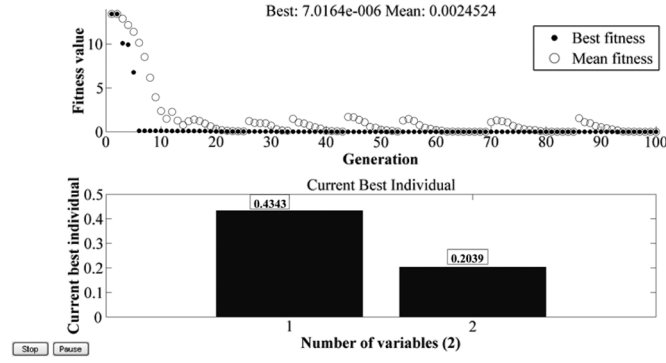


Fig. 5. Generation iteration process.

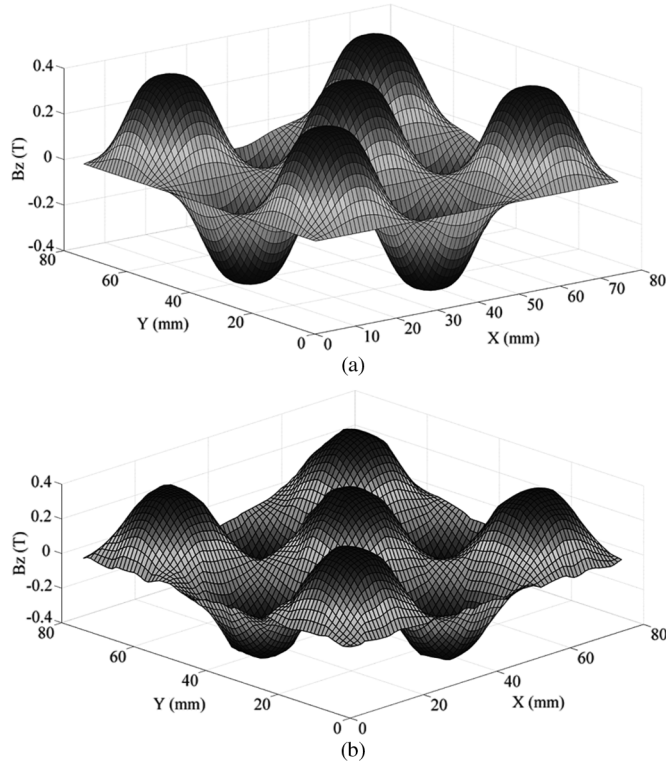


Fig. 6. Air-gap flux density distribution.

magnet surface when τ_m/τ takes 0.4343 and τ_n/τ takes 0.2039. Fig. 6(a) is the flux density calculated using (27). Fig. 6(b) is the flux density in the center area of proposed magnet array shown in Fig. 2(a), which is calculated by 3-D finite-element model. The edge effect of the model is taken into account. It can be found that Fig. 6(b) approximates to Fig. 6(a). The difference of the peak values between Fig. 6(a) and Fig. 6(b) is less than 0.011 T, which is 2.78% of the peak value of Fig. 6(a).

IV. SIMULATION AND COMPARISON

If τ_m/τ takes 0.68 and τ_n/τ takes 0, the proposed magnet array becomes to the Halbach array which Jansen used in [5]. The two magnet arrays, which have the same parameters shown in Table I, are compared in Table III.

In order to compare the two magnet arrays shown in Table III, sums for all the higher harmonic components ($k > 1$ or $l > 1$)

TABLE III
TWO MAGNET ARRAYS ARE COMPARED

Magnet array	τ_m/τ	τ_n/τ	$f(\tau_m/\tau, \tau_n/\tau)$
Proposed by this paper	0.4343	0.2039	0.0013%
Used by Jansen	0.68	0	6.2773%

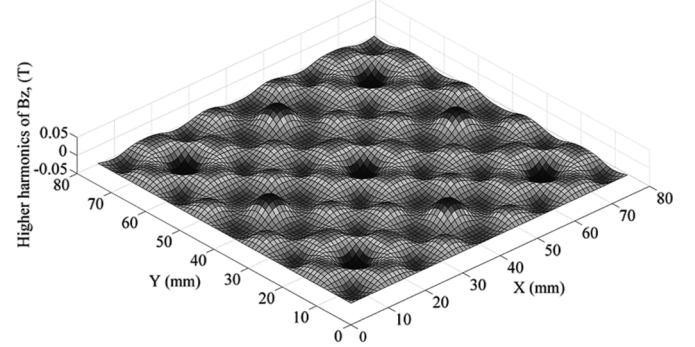


Fig. 7. The sum of the all the higher harmonic components of the magnet array proposed by this paper.

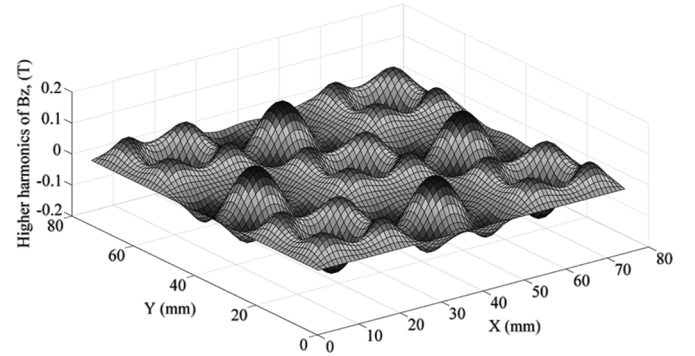


Fig. 8. The sum of the all the higher harmonic components of the magnet array used by Jansen.

of z -component of the magnetic flux density distribution on the x - y plane which is 4 mm apart from the magnet surface are gotten by harmonic model. The sums of the array proposed by this paper are shown in Fig. 7. The sums of the array used by Jansen are shown in Fig. 8. Comparing Fig. 7 and Fig. 8, it is found that higher harmonic components in the magnet array proposed by this paper are much smoother and smaller than the magnet array used by Jansen. The peak value of higher harmonic components in the magnet array proposed by this paper is about 37.77% of the magnet array used by Jansen. So the force ripple of the planar motor which uses the magnet array proposed by this paper will be smaller than planar motor which uses the magnet array used by Jansen.

Fig. 9 shows the peak value of z -component of the magnetic flux density along the air-gap length from the magnet surface. Fig. 10 shows the peak value of x -component of the magnetic flux density along the air-gap length from the magnet surface. It can be observed from Fig. 9 that the z -component of the magnetic flux density of the magnet array proposed by this paper is much larger than the array used by Jansen, which will generation more thrust force in x or y direction. In contrast, it is observed from Fig. 10 that the x -component of the magnetic flux density

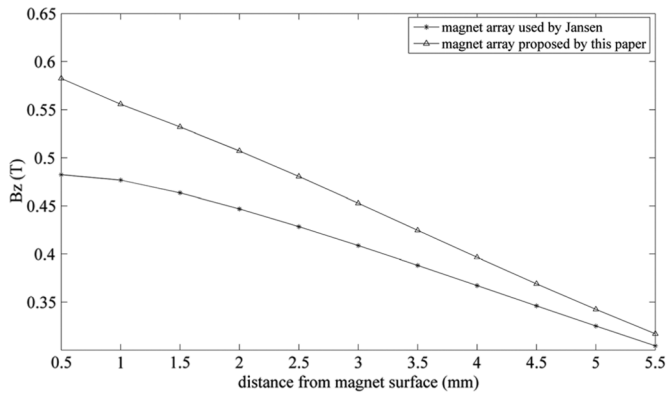


Fig. 9. Peak value of z -component of the magnetic flux density along air-gap length.

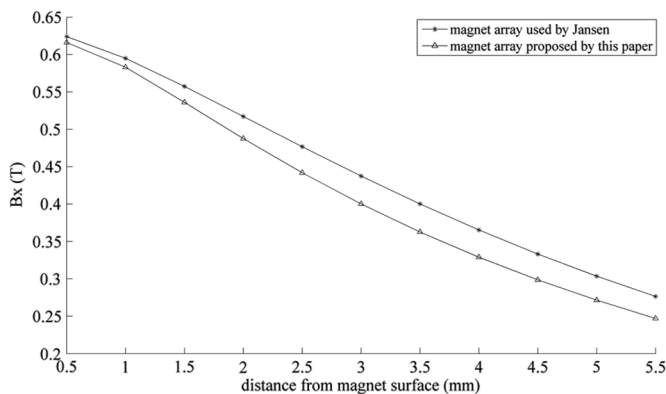


Fig. 10. Peak value of x -component of the magnetic flux density along air-gap length.

of the magnet array proposed by this paper is smaller than the array used by Jansen.

V. CONCLUSION

The higher harmonic components of air-gap flux density of magnet array are usually a main reason for the force ripples in the planar motor, especially since it uses air-core coils and has no iron yoke. To decrease higher harmonics components, a new magnet array is proposed, in which the harmonic model of the magnetic field was developed and validated by the finite-element method. The higher harmonic components of z -component of the magnetic flux density distribution for the proposed magnet array are minimized by a genetic algorithm. Compared with the magnet used by Jansen [5], it is found that:

- 1) The new magnet array has much less higher harmonics in z -component of the magnetic flux density distribution, which will decrease the force ripple of the planar motor.
- 2) The new magnet array has a much larger z -component of the magnetic flux density distribution, which may generate more thrust force in x or y direction in the planar motor.

- 3) The new magnet array has a smaller x -component (y component) of the magnetic flux density distribution, which may reduce the thrust force in z direction in the planar motor.

Unfortunately, the new magnet array is not investigated and optimized with other variables in the planar motor. Further studies on planar motors with the magnet array will be summarized in our next study.

ACKNOWLEDGMENT

This research was supported by the National Basic Research Program (973 Program) of China (2009CB724205), the National High Technology Research and Development Program ("863" Program) of China (2009AA04Z148), National Natural Science Foundation of China (No. 60904054), China Postdoctoral Science Foundation Funded Project (20080440384), and the independent research program (SKLT08B04) of the State Key Laboratory of Tribology of China.

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